

CLIENT HANDOVER

The FORGEDSA Wheel Configurator

What we built, how your customers use it, how you run it day to day, and the work still outstanding before launch. Written for business owners. No technical knowledge needed.

LIVE SITE	ADMIN	HANDED OVER
forged-configurator.vercel.app	forged-configurator.vercel.app/admin	28 September 2026

01

At a glance

A premium online 3D showroom. A BMW M owner picks their exact car and factory paint, then tries your forged wheels on it in real time: design, size and finish, from any angle, on a laptop or phone. They get a summary of their build with studio images and the price, a branded PDF, a share link and an enquiry form. Every PDF download and enquiry is emailed to you as a branded lead summary. Behind it is a secure back end and an admin panel, so day-to-day changes need no developer.

11
BMW M models, 9 with a real 3D body

64
BMW South Africa factory paints, matched per car

9
FORGEDSA wheel designs live (FSA-01 to FSA-09)

4
Finishes: Liquid Metal, Bronze Matte, Black Gloss, Silver

5
Regions and currencies, South Africa and Rand by default

<10s
From PDF download or enquiry to a lead email in your inbox

Where things stand

AREA	STATUS	NOTES
Customer journey	LIVE	Welcome → showroom → configurator → summary → PDF / enquiry / share, on desktop and mobile.
Brand	LIVE	FORGEDSA brand kit applied across the site, the PDF and the lead email.
Back end and admin	LIVE	Database, storage, secure logins, and an admin panel for cars, wheels, prices, finishes, paints and enquiries.
Leads	LIVE	PDF gated on email and mobile. Every lead is saved and emailed to forgedsa1@gmail.com.
Wheel fidelity	OUTSTANDING	Photo-based 3D today; true CAD-based 3D wheels next. See section 02.
Vehicle rendering	OUTSTANDING	Good at normal distance; body models and lighting to be raised to photoreal. See section 02.
Terms and Delivery	OUTSTANDING	Live as drafts; content to be checked for accuracy. See section 02.

02

Outstanding work

The configurator is live and working end to end. **Three pieces of work are outstanding** before we consider it finished. They are listed here first so nothing is missed.

OUTSTANDING · 1 OF 3

Improve the fidelity of the wheels

WHERE IT IS TODAY

- The nine live wheels (FSA-01 to FSA-09) are built from straight-on photos. The app turns each photo into 3D spokes.
- On the car, at normal viewing distance, they look convincing. Up close, and in the 360° view, spoke depth, the inner barrel and bolt detail are approximated.
- FSA-10 to FSA-17 only have angled photos, so they are not live yet.

WHAT WE WILL DO

- Build every design as a real 3D model from the manufacturer's CAD files: true spoke depth, barrel, lip, bolts and edges, photoreal from every angle.
- The route is already built. Uploading a CAD file in the admin panel converts it into a web-ready 3D wheel automatically, and it replaces the photo version on every car immediately.
- Bring FSA-10 to FSA-17 live the same way.

Needed from you: the CAD file for each design from your manufacturer (STEP/STP ideal; IGES, STL or OBJ also work), rim only, no tyre. One file per design.

OUTSTANDING · 2 OF 3

Improve the fidelity of the vehicle rendering

WHERE IT IS TODAY

- Nine cars have licensed community 3D bodies. Quality varies from model to model: the M5 is closest to photoreal, others show simpler interiors, lights and panel detail up close.
- Paint, reflections and lighting have been tuned for a realistic metallic and clearcoat look, but they are real-time approximations.
- The M4 Convertible and M5 Touring have no commercially licensed model yet, so they are hidden from the showroom.
- Several community models appear to derive from game assets, so a formal licence check is advisable before a large launch.

WHAT WE WILL DO

- Replace the weaker bodies with higher-detail, commercially licensed models (purchased where needed), prioritising your best-selling cars.
- Refine materials per car: glass, lights, tyres and brakes, badges, trim and paint break-up.
- Raise studio lighting, shadows and reflections, and render the summary and PDF stills at a higher quality than the live view.
- Source the M4 Convertible and M5 Touring, and confirm the licence of every model.

Needed from you: the priority order of models (which cars your customers drive most), and approval for any paid model purchases.

OUTSTANDING · 3 OF 3

Check the Terms & conditions and Delivery pages for accuracy

WHERE IT IS TODAY

- Both pages are live at **/terms** and **/delivery**, linked from the menu, the footer and the PDF.
- They are **drafts**: a sensible South African starting point (Consumer Protection Act, POPIA, deposits for made-to-order goods, cancellations, warranty, fitment, courier delivery, inspection on arrival).
- Each page shows a "Draft for review" note until approved.

WHAT NEEDS CHECKING

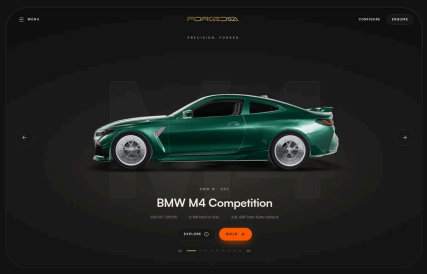
- Every statement matches how FORGEDSA actually trades: deposits, payment, lead times, delivery areas and costs, fitment, returns and warranty.
- Fill in all bracketed details: registered company name, registration and VAT numbers, deposit percentage, payment methods, warranty periods, lead times, delivery costs, collection and fitment address, Information Officer.
- Legal review by your attorney, then we update the text and remove the draft note.

Needed from you: your answers to the bracketed items and your attorney's sign-off. The change itself is quick: one file (config/pages.ts).

03

The customer journey

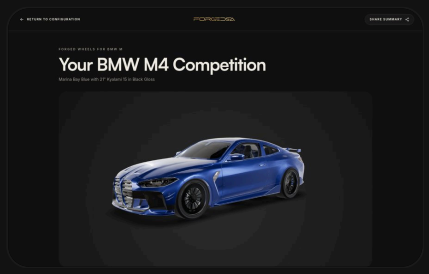
We mapped and built every step a customer takes, from the first second to the PDF they keep.



Showroom: one car at a time



Configurator: paint, wheels, size, finish



Summary: their build, price and actions

STEP	WHAT THE CUSTOMER SEES	WHY IT MATTERS
1 · Loading	A FORGEDSA wheel drawn in fine gold lines while the 3D studio loads.	On-brand from the first second.
2 · Welcome	A BMW M turning in a dark studio, the logo, a region and currency picker and **Enter**. Social links underneath.	Sets the tone and the right currency.
3 · Showroom	One car at a time, model name large behind it. Arrows drive the car out and the next in. **Explore** or **Build**.	Customers find *their* car.

4 · Notice	A one-line note that cars are visualisations and fitment is confirmed by your team.	Manages expectations.
5 · Configurator	Four steps: **Paint** → **Wheels** → **Size** → **Finish**. Wheel choices cut to a close-up. Running price always visible.	Their car on your wheels.
6 · 360° view	Any wheel on its own, rotating, with finishes, sizes and price.	Study the design up close.
7 · Summary	Three studio images, every option, the price, and Share, PDF and Enquire.	Turns interest into a lead.
8 · PDF	They enter **email and mobile** (name optional), then a two-page branded spec sheet downloads instantly.	Every download is a lead, emailed to you.
9 · Enquiry	Name, email, phone, message, "I'd like a consultation". Configuration attached automatically.	A qualified lead, by email in seconds.
10 · Share	A short code and link that reopens the exact build.	Easy to forward; you can open it too.

Also on the site: The collection (all wheel designs), Delivery & fitment, Terms & conditions, and a footer with your social channels and contact details.

What customers choose

Car	11 BMW M models: M2, M235i, M3, M3 Touring, M4, M4 Convertible, M5, M5 Touring, M8, X3 M50, X5 M (2024 facelift X5 M Competition).
Paint	The exact BMW South Africa factory colours for that car, 64 in total, researched from BMW's own configurator.
Wheel	Your real FORGEDSA designs: 9 live now.
Size	Only the sizes you offer for that wheel, each with its own price.
Finish	Liquid Metal (chrome), Bronze Matte, Black Gloss and Silver.

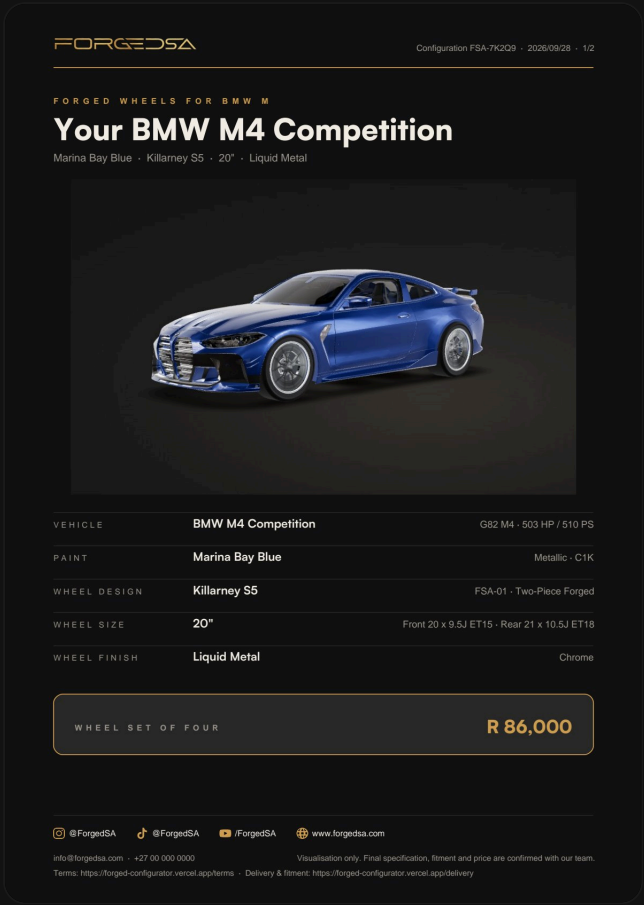
04

Your brand, applied

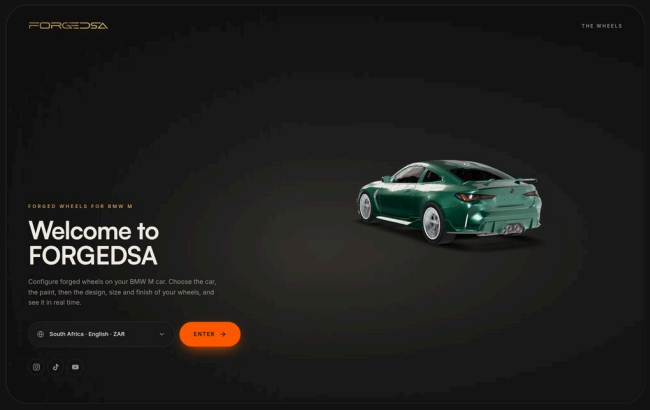
The whole experience follows the FORGEDSA Official Brand Design System: site, PDF brochure and lead email.

Matte Carbon	Gunmetal	Forged Gold	Brass Light	Electric Orange
#121212 · backgrounds	#2A2A2A · surfaces	#CF9F52 · logo, selections, prices	#E8C37D · hovers	#FF5A00 · main action buttons

- **Electric Orange** is the action colour. It is used only on the buttons that move a customer forward (Enter, Build, Confirm, View your build, Download PDF, Send enquiry), the "Open your build online" button in the PDF, and the Reply button in the lead email. **Forged Gold** marks everything else that stands out.
- **Type:** Satoshi for headlines, Inter for everything else, as the kit specifies. Both are hosted with the site, so they never fail to load.
- **Logo:** the gold FORGEDSA wordmark on every page, the PDF and the email. We cut it out of the supplied image, which had a printed checkerboard rather than real transparency.
- **Social channels:** Instagram @ForgedSA, TikTok @ForgedSA, YouTube /ForgedSA and www.forgedsa.com, on the welcome screen, menu, footer and PDF.
- **Browser tab icon:** a temporary gold "F" until we receive the hexagon monogram.


FORGEDSA
Configuration FSA-7K2Q9 · 2026/09/28 · 1/2
FORGED WHEELS FOR BMW M
Your BMW M4 Competition
Marina Bay Blue · Killarney S5 · 20" · Liquid Metal
VEHICLE BMW M4 Competition G82 M4 · 503 HP / 510 PS
PAINT Marina Bay Blue Metallic · C1K
WHEEL DESIGN Killarney S5 FSA-01 · Two-Piece Forged
WHEEL SIZE 20" Front 20 x 9.5J ET15 · Rear 21 x 10.5J ET18
WHEEL FINISH Liquid Metal Chrome
WHEEL SET OF FOUR R 86,000
@ForgedSA @ForgedSA /ForgedSA www.forgedsa.com
info@forgedsa.com · +27 00 000 0000
Visualisation only. Final specification, fitment and price are confirmed with our team.
Terms: https://forged-configurator.vercel.app/terms · Delivery & fitment: https://forged-configurator.vercel.app/delivery

The customer's PDF brochure, page 1



Welcome screen with the region picker and Enter

05

Behind the scenes

A proper back end keeps the site secure and lets you manage it without a developer.

PART	WHAT IT DOES	WHERE IT LIVES
Database	Every car, paint, wheel, finish and price, plus saved builds and customer enquiries.	Supabase
File storage	3D car files, wheel images and wheel 3D models.	Supabase Storage
Logins	Only people on your admin list can change anything, and the database itself enforces this.	Supabase Auth
Hosting	Serves the website worldwide, fast.	Vercel
Lead emails	Emails forgedsa1@gmail.com a branded summary of every PDF download and enquiry.	Database + n8n + Gmail
Wheel pipeline	Turns wheel photos plus a Google Sheet into wheels in the app automatically.	n8n + Google Sheet

Know-how written down: every recurring job (adding a car or wheels, tuning the paint, the lead emails, publishing changes, keeping keys safe) is written up as a step-by-step playbook inside the project (docs/SKILLS.md), so any developer or AI assistant can pick up the work without starting from scratch.

If the database were ever unreachable, the site falls back to a built-in copy of the catalogue, so a customer never sees an empty page.

06

Your admin panel

Go to **/admin** and sign in with your admin email and password.

First sign-in or forgotten password: on the sign-in screen click *First time here, or forgot your password?*, enter your admin email and press *Email me a link*. Open the email on the same device; it brings you back to choose your password. Your account (fazal@novahavens.com) already exists.

TAB	WHAT YOU CAN DO
Cars	Add a car, upload its 3D model, choose its factory paints, hide or reorder cars.
Wheels	Add a design, upload its image, 3D model or CAD file, write its copy, set **sizes and prices**, choose finishes, hide or reorder.
Finishes	Edit the four finishes: name, colour, gloss or matte.
Paints	Edit the factory paint list.
Enquiries	Every enquiry and PDF download, newest first: **Reply**, **Open build**, **Export CSV**.

Everyday tasks

Change a price	Wheels → the wheel → change the price next to a size → Save. Live immediately; the "from" price updates itself.
Add or remove a size	In the same wheel, use the + size buttons or the bin icon → Save.
Hide something	Untick *Visible in configurator* → Save. Nothing is deleted.
Change the order	Set *Sort order*: lower numbers appear first.
Add a car	Cars → New → name → upload the 3D file (.glb) → pick its paints → Save. A developer may fine-tune which parts take the paint the first time.
Add an admin	Create their user in Supabase (Authentication → Users → Add user, Auto Confirm) and add their email to the admin list. They then use the "First time here" link.

07

New-lead emails

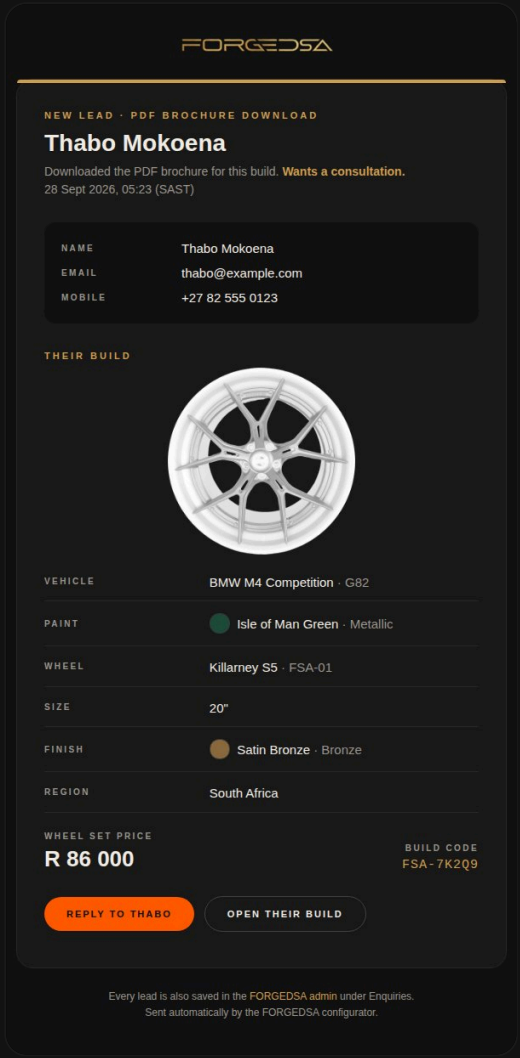
Every time a customer **downloads the PDF** or **sends an enquiry**, a branded email lands in **forgedsa1@gmail.com** within seconds.

What it shows

- The subject says the kind of lead: *"New PDF download: ..."* or *"New enquiry: ..."*, with the name and car.
- Name, email and mobile (tap to email or call), when they came in, and whether they asked for a consultation.
- Their message, for enquiries.
- **Their build:** wheel image, car, paint with swatch, wheel and code, size, finish with swatch, region.
- The wheel set price in their currency, and their build code.
- **Reply to [name]** and **Open their build** buttons. Pressing Reply in Gmail also goes straight to the customer.

How it works

- The lead is saved in the database (the Enquiries tab), as before.
- The moment it is saved, the database hands it to the n8n automation *FORGEDSA · Lead email*.
- n8n builds the email and sends it through the connected Gmail account (currently fazal@novahavens.com, shown as "FORGEDSA Leads").



A lead email (sample details)

If an email ever fails, the lead is **still saved** in the admin panel. Nothing is lost, and the customer never sees an error.

Good to know

- **Change the recipient:** n8n → the workflow → *Email forgedsa1@gmail.com* step → To → Publish.
- **Send from another account** (e.g. forgedsa1@gmail.com itself): connect it as a Gmail credential in n8n and select it in that step.
- **Check it is working:** n8n → the workflow → Executions.
- If Gmail files the first emails under Promotions, drag one to Primary.

08

Your wheels

Live today

Nine FORGEDSA designs are live (FSA-01 to FSA-09), made from the front-on photos in your *Wheel designs* zip. The names (South African circuits), codes and prices are **placeholders** so you could see the experience end to end. Change them in the admin panel or the wheels sheet.

The wheels sheet

An editable Google Sheet called **wheels** sits in your FORGEDSA Drive folder: one row per design, with photo file name, name, code, sizes, prices in Rand, finishes and product copy. The automated pipeline reads it to add new wheels. Rows marked *skip = yes* are left alone.

The photoreal route: CAD files → real 3D wheels

Forged wheels are machined from CAD drawings, so the exact 3D shape of every design already exists at your manufacturer. This is how the wheel-fidelity work in section 02 gets done:

1 · Request	Ask your manufacturer for the CAD file of each design (STEP ideal; IGES, STL, OBJ fine). Rim only, no tyre.
2 · Upload	Admin → Wheels → the wheel → *Rim 3D model* → Upload the CAD file.
3 · Automatic	It is converted into a light, web-ready 3D wheel: turned to face outwards, sized, smoothed, and given one finish surface so all four finishes apply.
4 · Save	A 3D model always takes priority over the photo, so the wheel upgrades on every car immediately.

Tips: if the spokes ever face into the car, tick *Flip 3D rim* and Save. Very large files (over about 50 MB) are better converted by your developer with the included converter, which does the same thing on a computer.

09

Details to send us before launch

Besides the three pieces of outstanding work, these details replace placeholders on the site.

- ☐ **CAD files** for each wheel design (for outstanding work 1).
- ☐ **Car priority and approval** for any paid 3D model purchases (for outstanding work 2).
- ☐ **Bracketed Terms and Delivery details** and your attorney's sign-off (for outstanding work 3).
- ☐ Real wheel **names, codes, sizes and prices**. The current ones are placeholders.
- ☐ Your real **email address and phone number**. The site shows info@forgedsa.com and +27 00 000 0000.
- ☐ The **hexagon monogram** (PNG or SVG with transparency) for the browser tab icon.
- ☐ **Wheel weights**, if you want them shown. They appear automatically once entered.
- ☐ Your own **domain** (e.g. configure.forgedsa.com), if wanted. We can point it at the site in minutes.

10

Accounts and running costs

SERVICE	USED FOR	PLAN
Vercel	Hosting the website	Free tier; upgrade if traffic grows.
Supabase	Database, storage, logins	Free tier. Free projects pause after a week with no visits: open the site weekly, or upgrade to Pro (about US$25/month) at launch.
n8n	Wheel pipeline and lead emails	Your existing account.
Gmail	Sending lead emails	Free.
Google Drive	Wheels sheet and photos	Your existing account.
OpenRouter	AI image tools in the wheel pipeline	Pay per use, about US$0.15–0.30 per wheel.

11

Quick glossary

Back end	The behind-the-scenes part (database, storage, logins) customers never see.
Admin panel	Your private page for editing the catalogue and reading enquiries.
3D model / GLB	A file describing an object's shape in 3D, like the cars and your wheels.
CAD file / STEP	The engineering drawing your manufacturer machines the wheel from.
Fidelity	How closely the on-screen image matches the real thing.
Build code	The short code that reopens a customer's exact build.
Pipeline	An automation that turns photos plus a spreadsheet into wheels in the app.
Webhook	A private web address one system calls to hand information to another.

The technical companion for a developer is docs/HANDOVER.md in the project repository.